

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)			Order is: B, C, A, D. 3 marks for all 4 correct 2 marks for 2 or 3 correct 1 mark for 1 correct	3			3		3
	(b)			Any 2 × (1) from: - Move the ruler nearer [the spring] - Clamp the ruler in place / ensure the ruler is vertical - Use a ruler marked in mm or with better resolution. Accept measure in mm. - Use a pointer [from the (bottom of the) spring to the ruler] Don't accept use a laser pointer. Don't accept repeat the experiment or use a metre ruler or use a camera or more accurate ruler.			2	2		2
	(c)	(i)		4 [N]	1			1	1	1
		(ii)		$k = \frac{4 \text{ ecf}}{8} (1)$ = 0.5 [N/cm] (1)	1	1		2	2	2
		(iii)		400 g to 200 g (1) So 8 cm goes to 4 cm (1) [so the claim is false]. Alternative: Mass is half of 400 g (1) So the extension is 4 cm (1) [so the claim is false]. Alternative: 2 cm is a quarter of the 8 cm extension (1) so the force is 1 N / mass of 100 g (1) [so the claim is false].			2	2	1	2
				Question 3 total	5	1	4	10	4	10